$$\textit{Pearl's Vital Index} = \frac{CBR}{CDR} \times 100$$

Positive values of Crude Rate of Natural Increase indicate the net increase in the population. Similarly, negative values of Crude Rate of Natural Increase indicate the net decrease in the population.

If Pearl's Vital Index is greater than 100, then it can be regarded as the population is growing. On the other hand, if this index is less than 100, it can be regarded as the population is not growing. The above formula shows that the Vital Index can also provide knowledge on birth-death ratio of the population.

These two measures are simple and easy to calculate. They indicate whether the number of births exceeds the number of deaths. However, these two measures suffer from the limitations of *CBR* and *CDR*. They cannot be used for comparing two different populations. Also, information regarding whether the population has a tendency to increase or decrease cannot be obtained from these two measures.

8.2 OFFICIAL STATISTICS

Official statistics are the statistical information published by government agencies or other public bodies, which are collected and compiled on various aspects for administrative purposes. Official Statistics are collected in a systematic manner through a well-established Statistical System. These information include quantitative and qualitative information on all major areas of citizens' lives, such as economic and social development, living conditions, health, education and environment. Official statistics should be objective and easily accessible, produced on a continual basis so that measurement of change is possible. The following may be considered as the main functions of the Statistical System:

(i) collection, validation, compilation of data

(ii) publication/dissemination of the statistical information

(iii) maintenance of statistical standards such as definitions, classification, statistical methodology, comparability *etc*.

(iv) coordination of statistical activities

(v) training statistical personnel

(vi) independence and integrity of its functioning

(vii) international coordination

8.2.1 Early History of Statistical System in India

Statistical data collection and compilation began in India during 321-298 *BC* and are documented in Kautilya's *Arthasastra*. Later, during the Moghul's period, the details of Official Statistics can be found in *Ain-i-Akbari* written by Abul Fazal in Emperor Akbar's rule during 1590 *AD*. It contains Official Statistics of various characteristics including land classification, crop yields, measurement system, revenue *etc*.

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